LangChain-M-Set1

Applied composition and retrieval | 20 MCQs | Choose one best answer

Python snippets assume standard imports and the stated setup.

01. The input is a dictionary, and the prompt expects question to be a string. retriever accepts a string; format_docs accepts its returned documents. Which minimal edit fixes the question field without changing retrieval?

```
chain = {
    "context": itemgetter("question") | retriever | format_docs,
    "question": RunnablePassthrough(),
} | prompt | model
chain.invoke({"question": "What changed?"})
```

A. Replace format_docs with StrOutputParser().

B. Replace the whole mapping with RunnablePassthrough().

C. Replace the question branch with itemgetter("question").

D. Change the input to the chain into a bare string.

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02. The input contains only amount=100. The second computed field must use the newly calculated tax, not an existing input field. Which change makes the intended total=110 reliable?

```
step = RunnablePassthrough.assign(
    tax=lambda x: x["amount"] * 0.10,
    total=lambda x: x["amount"] + x["tax"],
)
```

A. Use one assign for tax, followed by a second assign for total.

B. Keep one assign and place the tax keyword before total.

C. Keep one assign and add a larger max_concurrency value.

D. Replace assign with a parallel mapping that reads input["tax"].

---

03. Assume standard RunnableLambda imports. What does this expression return, and what data flow explains it?

```
chain = RunnableLambda(lambda x: x * 2) | {
    "a": RunnableLambda(lambda x: x + 1),
    "b": RunnableLambda(lambda x: x * 2),
}
chain.invoke(3)
```

A. 14; the mapping is shorthand for sequentially applying a then b.

B. {"a": 7, "b": 14}; branch b receives branch a's result.

C. {"a": 4, "b": 6}; both branches receive the original 3.

D. {"a": 7, "b": 12}; both branches receive 6.

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04. model streams tokens, but clients receive no text until the response finishes after a full-string normalization function is appended. Retrieval latency is unchanged. Which explanation and remedy fit best?

```
chain = prompt | model | StrOutputParser()
chain = chain | RunnableLambda(normalize_complete_text)
```

A. The model has stopped supporting streaming; replace the provider before inspecting the transform.

B. The full-input transform buffers upstream output; use a chunk-aware transform if normalization permits it.

C. The normalizer needs more parallel workers; increase batch concurrency to release each token.

D. The client must call invoke instead of stream so normalization can run incrementally.

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05. A labeling job sends three documents to batch_as_completed. They finish in the order document 2, document 0, document 1. Which consumer preserves the correct document-to-label association?

A. Use the returned input index to attach each result to its original document.

B. Zip the original document list with the completion stream as it arrives.

C. Sort labels alphabetically, then zip them with the original document list.

D. Use the result arrival position as the original document index.

---

06. A developer invokes model.bind_tools([lookup_invoice]) and receives an AIMessage containing a tool call. No invoice lookup is recorded. The requirement is to execute permitted tool calls and let the model use their results. What is missing?

A. A lower temperature so the tool function executes during text generation.

B. A second bind_tools call with the identical tool list.

C. A tool execution loop, such as create_agent or an explicitly implemented dispatcher.

D. A string parser after the model to execute the function name in the response.

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07. An extractor returns a Pydantic object with revenue=125.0 and currency="USD". It validates successfully, but the source says the figures are in thousands of EUR. What control most directly addresses this defect?

A. Add a second JSON parser after Pydantic validation without checking the source.

B. Change float to str so the extractor can preserve any wording it prefers.

C. Treat successful schema validation as sufficient and suppress the source warning.

D. Check extracted values, units, and currency against supporting source spans.

08. An approved model supports tool calling but lacks native schema-enforced responses. A create_agent application needs a validated Pydantic extraction result. Which configuration is the appropriate supported starting point?

A. Force ProviderStrategy(schema) despite the missing provider capability.

B. Use ToolStrategy(schema) and handle structured-output validation failures.

C. Use response_format=None and assume the final message is a Pydantic object.

D. Expose the schema only in a tool docstring and skip result validation.

09. history is a list of HumanMessage and AIMessage objects. The requirement is to preserve them as separate conversational turns. Which edit best fixes the prompt construction below?

```
prompt = ChatPromptTemplate.from_messages([
    ("system", "Answer using the supplied evidence."),
    ("human", "{history}"),
    ("human", "{question}"),
])
```

A. Replace it with ("system", "{history}") to increase the history's priority.

B. Replace ("human", "{history}") with MessagesPlaceholder("history").

C. Convert history to one JSON string and retain the human template.

D. Insert history into the current question string before formatting.

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10. For a comparison question, a query rewriter produces short keyword searches. Retrieval improves, but answers now address only the keywords and omit the requested comparison. Which data contract best fixes the pipeline?

A. Retain the original question for generation and use rewritten queries only for retrieval.

B. Replace the original question everywhere with the shortest rewritten query.

C. Remove query rewriting and compensate only by raising model temperature.

D. Concatenate all query variants and treat them as new user instructions.

11. An index was built with embedding model A. A deployment changes only query embeddings to model B. Both produce 1,536 dimensions, yet retrieval deteriorates. Which diagnosis is most defensible?

A. Increase k indefinitely because more results mathematically repair incompatible vector spaces.

B. Equal dimensionality guarantees compatibility; the reranker must be the only possible cause.

C. Equal dimensionality does not imply aligned spaces; use a compatible query/document embedding pair.

D. Pad both vectors to a larger dimension and keep the mixed embedding models.

12. Retrieved Documents contain source_id and page in metadata. A formatter keeps only page_content. The model later invents page numbers in citations. Which change makes citations auditable?

A. Generate page numbers from each chunk's rank in the retrieval result.

B. Ask the model more strongly to infer page numbers from the writing style.

C. Assign every chunk the same citation ID so duplicate text cannot create conflicts.

D. Preserve stable evidence IDs and a source/page mapping, then validate citations against that mapping.

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13. A query asks for several distinct causes. The top similarity hits are near-duplicate chunks about one cause. The vector store supports MMR. Which experiment directly targets this redundancy?

A. Lower the similarity score of every document equally and keep the same ranking.

B. Keep identical similarity results and ask the generator to invent the missing causes.

C. Set fetch_k below k so MMR has fewer alternatives to compare.

D. Retrieve a broader candidate pool, then use MMR to balance relevance and diversity for the final k.

14. A FAISS integration is explicitly configured to return squared L2 distances. Scores are A=0.12, B=0.65, C=1.40. A developer keeps scores >=0.80 as "high confidence." What is the immediate error?

A. Lower distance means closer here; the filter retains the weakest of these matches.

B. Higher distance always means higher semantic relevance, so the filter is correct.

C. The three scores are calibrated probabilities and must sum to one.

D. L2 distances become similarity probabilities when accessed through a retriever.

15. A PDF policy table is split into fixed character chunks. Retrieved cells often lack their column headers, causing unit mistakes. What is the best targeted ingestion change?

A. Increase chunk overlap blindly until every chunk nearly duplicates the document.

B. Preserve table/header relationships in chunks and retain document provenance.

C. Remove headers during ingestion because they are repetitive retrieval noise.

D. Keep the chunks and let the language model infer missing units from nearby numbers.

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16. For a failed query, the required passage is absent from the 20 documents passed to a reranker. The reranker selects 5 of those 20 for generation. Which change should be investigated first?

A. Increase generation temperature so the missing passage can be reconstructed.

B. Increase the reranker's output from 5 to 10 while keeping the same 20 candidates.

C. Improve candidate retrieval so the required passage can enter the reranker's input set.

D. Change the display order of the five selected citations without altering retrieval.

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17. An agent confuses search_current_policy with search_archived_policy. Their descriptions both say "Search policies," and the date argument is an unqualified string. Which change most directly improves selection and argument quality?

A. Repeat the same generic description several times in the system prompt.

B. Rename both tools to shorter opaque labels to reduce prompt tokens.

C. Define distinct tool purposes, temporal applicability, and validated date/identifier fields.

D. Add more nearly identical policy tools without changing the existing schemas.

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18. An async web endpoint calls a synchronous, network-blocking chain.invoke directly. The model provider has a supported async integration. Under load, unrelated requests stall. Which change best addresses event-loop blocking?

A. Keep chain.invoke and increase the number of tokens generated per request.

B. Await chain.ainvoke and audit custom stages for blocking calls; offload unavoidable synchronous work.

C. Call chain.ainvoke without awaiting or scheduling the returned coroutine.

D. Replace every async handler with a busy loop that waits for chain.invoke to finish.

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19. A chain ends with StrOutputParser. The model emits the text {"approved": false}. A downstream service expects a mapping with a Boolean approved field. What is the appropriate boundary change?

A. Search for the substring "approved" and treat its presence as True.

B. Keep StrOutputParser because valid-looking JSON text is already a Python dictionary.

C. Cast the entire string with bool(...) and use that as the approval decision.

D. Use structured output or JSON parsing plus schema validation before calling the service.

20. A narrowly scoped internal Q&A product must consult the approved knowledge base on every request before answering. Discretionary tool choice adds latency and occasionally skips search. Which architecture best satisfies the stated requirement?

A. Run a fixed retrieval-then-generation chain, with an explicit insufficient-evidence path.

B. Ask an agent to decide whether retrieval is useful, using a stronger optional-search prompt.

C. Answer first, then retrieve documents only to add citations to the existing answer.

D. Use an unconstrained multi-tool agent and count any tool invocation as evidence retrieval.